

MH2500 Probability and Introduction to Statistics

Slides for Week 8-1

Topics: Finding pdf

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Example

Let X be uniformly distributed over $(0, 1)$. We obtain the distribution of the random variable Y , defined by $Y = X^n$, as follows:

- For $y \leq 0$,

$$F_Y(y) = P\{Y \leq y\} \leq P\{Y \leq 0\} = P\{X^n \leq 0\} \leq P\{X \leq 0\} = 0.$$

- For $0 \leq y \leq 1$,

$$F_Y(y) = P\{Y \leq y\} = P\{X^n \leq y\} = P\{X \leq y^{1/n}\} = F_X(y^{1/n}) = y^{1/n},$$

as X is uniform on $[0, 1]$.

- For $y \geq 1$, $F_Y(y) \geq P\{Y \leq y\} \geq P\{Y \leq 1\} = 1.$

We then obtain the density function of Y **by differentiation**:

$$f_Y(y) = \begin{cases} \frac{1}{n}y^{\frac{1}{n}-1} & \text{if } 0 \leq y \leq 1; \\ 0 & \text{otherwise.} \end{cases}$$

Example

If X is a continuous random variable with probability density f_X , then the distribution of $Y = X^2$ is obtained as follows:

- For $y \leq 0$, $F_Y(y) = P\{Y \leq y\} = P\{X^2 \leq y\} \leq P\{X^2 \leq 0\} = 0$.
- For $y \geq 0$, $F_Y(y) = P\{Y \leq y\} = P\{X^2 \leq y\}$
$$= P\{-\sqrt{y} \leq X \leq \sqrt{y}\} = F_X(\sqrt{y}) - F_X(-\sqrt{y}).$$

We then obtain the density function of Y by differentiation:

$$\begin{aligned}f_Y(y) &= (F_Y(y))' = (F_X(\sqrt{y}) - F_X(-\sqrt{y}))' \\&= f_X(\sqrt{y})(\sqrt{y})' - f_X(-\sqrt{y})(-\sqrt{y})' \\&= f_X(\sqrt{y})\frac{1}{2\sqrt{y}} - f_X(-\sqrt{y})\left(-\frac{1}{2\sqrt{y}}\right) \\&= \frac{1}{2\sqrt{y}} [f_X(\sqrt{y}) + f_X(-\sqrt{y})].\end{aligned}$$

Proposition

Let X be a continuous random variable with probability density $f_X(x)$.

- As X is a continuous random variable, it is a continuous function from the sample space Ω to an interval $I = (a, b)$.
- I can be finite or infinite.
- $f_X(x) = 0$ if x is not in I .

Let $Y = g(X)$ where g is a differentiable (hence continuous), strictly monotonic function on $\mathbb{R}$. Then Y has the density function

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| & \text{if } y = g(x) \text{ for some } x \in I, \\ 0 & \text{otherwise.} \end{cases}$$

Here, as g is strictly monotonic, g^{-1} , the inverse function of g , exists:

$$g^{-1}(y) = x \text{ if } y = g(x).$$

Proof: We prove for the case when g is an increasing function. Then, as g^{-1} is nondecreasing, its derivative is nonnegative:

$$\left| \frac{d}{dy} g^{-1}(y) \right| = \frac{d}{dy} g^{-1}(y).$$

Let $Y = g(X)$. For $y = g(x)$ for some $x \in I$, $g^{-1}(y) = x \in I$ and

$$\begin{aligned} F_Y(y) &= P\{g(X) \leq y\} = P\{\omega \in \Omega : g(X(\omega)) \leq y\} \\ &= P\{\omega \in \Omega : X(\omega) \leq g^{-1}(y)\} = F_X(g^{-1}(y)). \end{aligned}$$

By differentiation (chain's rule), we have

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y).$$

- For $y \neq g(x)$ for any $x \in I$, then $g^{-1}(y)$ is either less than a or bigger than b , and $F_Y(y)$ is either 0 or 1. In both cases, $f_Y(y) = 0$.

Example:

Let X be a continuous **nonnegative** random variable with density function f , and let $Y = X^n$.

- Find f_Y , the probability density function of Y .

Solution: For $g(x) = x^n$, a differentiable, strictly increasing function of x ,

$$g^{-1}(y) = y^{\frac{1}{n}}, \quad \frac{d}{dy}\{g^{-1}(y)\} = \frac{1}{n}y^{1/n-1}.$$

By **proposition above**, for $y \geq 0$,

$$f_Y(y) = \frac{1}{n}y^{1/n-1}f(y^{1/n}).$$

- When $n = 2$, $Y = X^2$ and $f_Y(y) = \frac{1}{2\sqrt{y}}f(\sqrt{y})$.
- Compare this with the one we obtained before the proposition.

Proposition

Let X be a continuous random variable with CDF F , and let $Z = F(X)$. Then Z has a uniform distribution on $[0, 1]$.

Proof: Note that the cumulative distribution function is increasing (not necessary to be strictly increasing) of x and has range $[0, 1]$.

For $z \in [0, 1]$, note that $F^{-1}(z)$ is a number and

$$\begin{aligned} P\{Z \leq z\} &= P\{F(X) \leq z\} \\ &= P\{\omega \in \Omega : F(X(\omega)) \leq z\} \\ &= P\{\omega \in \Omega : X(\omega) \leq F^{-1}(z)\} \\ &= P\{X \leq F^{-1}(z)\} = F(F^{-1}(z)) = z. \end{aligned}$$

This shows that $f_Z(z) = 1$ for all $z \in [0, 1]$. Also we have for $z < 0$, $P\{Z \leq z\} = 0$ and for $z > 1$, $P\{Z \leq z\} = 1$, which gives $f_Z(z) = 0$ for $z \notin [0, 1]$, and hence Z has a uniform distribution on $[0, 1]$.

Summary

We have studied continuous random variables, probability density function, cumulative distribution function, expectation and variance.

We focused on the following distributions:

- Uniform distribution
- Normal distribution and standard normal distribution
 - Approximating Binomial distribution by using normal distribution
- Exponential distribution
- Gamma distribution

We should know the parameters involved in distribution, expectation and variance, and tell which distribution can be applied for given questions.